

The Ephemeris

June 2021

Volume 32 Number 02- The Official Publication of the San Jose Astronomical Association



June 2021 - September 2021 Events

Schedule subject to change due to Covid-19 restrictions.

Please check our website for up to date information.

Board & General Meetings

Saturday 6/26, 7/24, 8/21, 9/18

Board Meetings: 6 -7:30pm

General Meetings: 8-9:30pm

Fix-It Day (2-4pm)

Currently on hold due to COVID-19 Situation

Solar Sunday (2-4pm)

Sunday 6/6, 7/11, 8/2, 9/13

Intro to the Night Sky Class & Houge Park 1Q In-Town Star Party

Currently on hold due to COVID-19 Situation

Houge Park 3Q In-Town Star Party

Currently on hold due to COVID-19 Situation

RCDO Starry Nights Star Party

Currently on hold due to COVID-19 Situation

Imaging SIG Mtg

Tuesday 6/15, 7/20, 8/17, 9/21

Astro Imaging Workshops (at Little Uvas OSP)

Currently on hold due to COVID-19 Situation

Binocular Star Gazing

Saturday (will resume Summer 2021)

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

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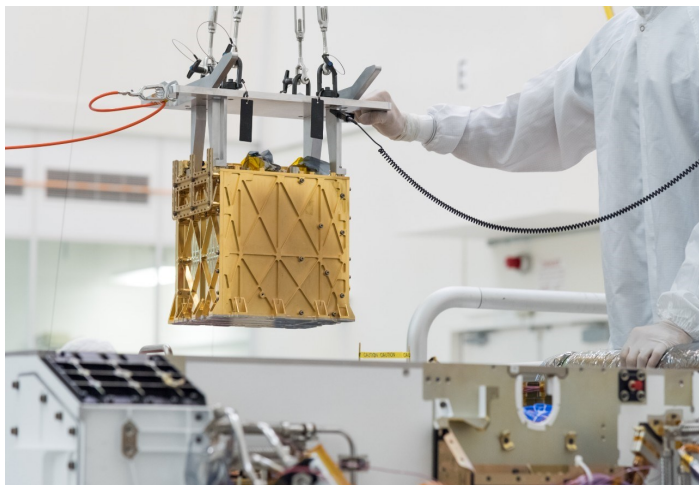
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Statue of Liberty Nebula
Photo by SJAA Member: Glenn Newell
May 25, 2021
Location: Chile

Glenn C. Newell Copyright 2021

NASA's Perseverance Mars Rover Extracts First Oxygen from Red Planet

Credit: NASA



Technicians at NASA's Jet Propulsion Laboratory lower the Mars Oxygen In-Situ Resource Utilization Experiment (MOXIE) instrument into the belly of the Perseverance rover.

Credits: NASA/JPL-Caltech

The growing list of “firsts” for Perseverance, NASA's newest six-wheeled robot on the Martian surface, includes converting some of the Red Planet's thin, carbon dioxide-rich atmosphere into oxygen. A toaster-size, experimental instrument aboard Perseverance called the Mars Oxygen In-Situ Resource Utilization Experiment (MOXIE) accomplished the task. The test took place April 20, the 60th Martian day, or sol, since the mission landed Feb. 18.

While the technology demonstration is just getting started, it could pave the way for science fiction to become science fact – isolating and storing oxygen on Mars to help power rockets that could lift astronauts off the planet's surface. Such devices also might one day provide breathable air for astronauts themselves. MOXIE is an exploration technology investigation – as is the Mars Environmental Dynamics Analyzer (MEDA) weather station – and is sponsored by NASA's Space Technology Mission Directorate (STMD) and Human Exploration and Operations Mission Directorate.

“This is a critical first step at converting carbon dioxide to oxygen on Mars,” said Jim Reuter, associate administrator for STMD. “MOXIE has more work to do, but the results from this technology demonstration are full of promise as we move toward our goal of one day seeing humans on Mars. Oxygen isn't just the stuff we breathe. Rocket propellant depends on oxygen, and future explorers will depend on producing propellant on Mars to make the trip home.”

For rockets or astronauts, oxygen is key, said MOXIE's principal investigator, Michael Hecht of the Massachusetts Institute of Technology's Haystack Observatory.

To burn its fuel, a rocket must have more oxygen by weight. Getting four astronauts off the Martian surface on a future mission would require approximately 15,000 pounds (7 metric tons) of rocket fuel and 55,000 pounds (25 metric tons) of oxygen. In contrast, astronauts living and working on Mars would require far less oxygen to breathe. “The astronauts who spend a year on the surface will maybe use one metric ton between them,” Hecht said.

Hauling 25 metric tons of oxygen from Earth to Mars would be an arduous task. Transporting a one-ton oxygen converter – a larger, more powerful descendant of MOXIE that could produce those 25 tons – would be far more economical and practical.

Mars' atmosphere is 96% carbon dioxide. MOXIE works by separating oxygen atoms from carbon dioxide molecules, which are made up of one carbon atom and two oxygen atoms. A waste product, carbon monoxide, is emitted into the Martian atmosphere.

The conversion process requires high levels of heat to reach a temperature of approximately 1,470 degrees Fahrenheit (800 Celsius). To accommodate this, the MOXIE unit is made with heat-tolerant materials. These include 3D-printed nickel alloy parts, which heat and cool the gases flowing through it, and a lightweight aerogel that helps hold in the heat. A thin gold coating on the outside of MOXIE reflects infrared heat, keeping it from radiating outward and potentially damaging other parts of Perseverance.

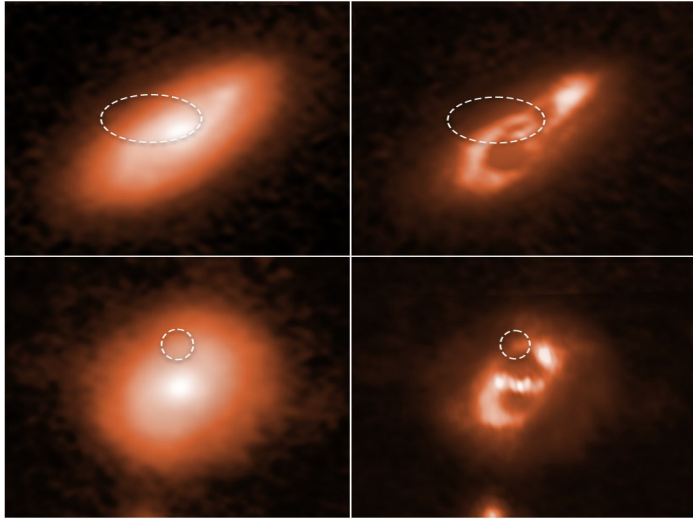
More About Perseverance

A key objective of Perseverance's mission on Mars is astrobiology, including the search for signs of ancient microbial life. The rover will characterize the planet's geology and past climate, pave the way for human exploration of the Red Planet, and be the first mission to collect and cache Martian rock and regolith (broken rock and dust).

Subsequent NASA missions, in cooperation with ESA (European Space Agency), would send spacecraft to Mars to collect these sealed samples from the surface and return them to Earth for in-depth analysis.

Hubble Tracks Down Fast Radio Bursts to Galaxies' Spinal Nuts

Credit: NASA



Astronomers using the Hubble Space Telescope have tracked down two brief, powerful radio bursts to the spiral arms of the two galaxies shown above. The two images at left show the full Hubble snapshots of each galaxy. The two digitally enhanced images on the right reveal each galaxy's spiral structure in more detail. The catalogue names of the bursts are FRB 190714 (top row), and FRB 180924 (bottom row). The galaxies are far from Earth, appearing as they looked billions of years ago. The dotted oval lines in each of the four images mark the location of the brilliant radio flares.

Credits: NASA, ESA, Alexandra Mannings (UC Santa Cruz), Wen-Fai Fong (Northwestern) IMAGE PROCESSING: Alyssa Pagan (STScI)

Astronomers using NASA's Hubble Space Telescope have traced the locations of five brief, powerful radio blasts to the spiral arms of five distant galaxies. Called fast radio bursts (FRBs), these extraordinary events generate as much energy in a thousandth of a second as the Sun does in a year. Because these transient radio pulses disappear in much less than the blink of an eye, researchers have had a hard time tracking down where they come from, much less determining what kind of object or objects is causing them. Therefore, most of the time, astronomers don't know exactly where to look. Locating where these blasts are coming from, and in particular, what galaxies they originate from, is important in determining what kinds of astronomical events trigger such intense flashes of energy. The new Hubble survey of eight FRBs helps researchers narrow the list of possible FRB sources.

The first FRB was discovered in archived data recorded by the Parkes radio observatory on July 24, 2001. Since then astronomers have uncovered up to 1,000 FRBs, but they have only been able to associate roughly 15 of them to particular galaxies.

"Our results are new and exciting. This is the first high-resolution view of a population of FRBs, and Hubble reveals that five of them are localized near or on a galaxy's spiral arms," said Alexandra Mannings of the University of California, Santa Cruz, the study's lead author. "Most of the galaxies are massive, relatively young, and still forming stars. The imaging allows us to get a better idea of the overall host-galaxy properties, such as its mass and star-formation rate, as well as probe what's happening right at the FRB position because Hubble has such great resolution." In the Hubble study, astronomers not only pinned all of them to host galaxies, but they also identified the kinds of locations they originated from. Hubble observed one of the FRB locations in 2017 and the other seven in 2019 and 2020. "We don't know what causes FRBs, so it's really important to use context when we have it," said team member Wen-Fai Fong of Northwestern University in Evanston, Illinois. "This technique has worked very well for identifying the progenitors of other types of transients, such as supernovae and gamma-ray bursts. Hubble played a big role in those studies, too." The galaxies in the Hubble study existed billions of years ago. Astronomers, therefore, are seeing the galaxies as they appeared when the universe was about half its current age.

Many of them are as massive as our Milky Way. The observations were made in ultraviolet and near-infrared light with Hubble's Wide Field Camera 3.

Location, Location, Location

The images display a diversity of spiral-arm structure, from tightly wound to more diffuse, revealing how the stars are distributed along these prominent features. A galaxy's spiral arms trace the distribution of young, massive stars. However, the Hubble images reveal that the FRBs found near the spiral arms do not come from the very brightest regions, which blaze with the light from hefty stars. The images help support a picture that the FRBs likely do not originate from the youngest, most massive stars.

These clues helped the researchers rule out some of the possible triggers of types of these brilliant flares, including the explosive deaths of the youngest, most massive stars, which generate gamma-ray bursts and some types of supernovae. Another unlikely source is the merger of neutron stars, the crushed cores of stars that end their lives in supernova explosions. These mergers take billions of years to occur and are usually found far from the spiral arms of older galaxies that are no longer forming stars.

Crew-1 Astronauts Safely Splash Down After Space Station Mission

Credit: NASA



NASA astronauts Shannon Walker, left, Victor Glover, Mike Hopkins, and Japan Aerospace Exploration Agency (JAXA) astronaut Soichi Noguchi, right are seen inside the SpaceX Crew Dragon Resilience spacecraft onboard the SpaceX GO Navigator recovery ship shortly after landing in the Gulf of Mexico off the coast of Panama City, Florida, at 2:56 a.m. EDT May 2, 2021.

Credits: NASA/Bill Ingalls

Four astronauts splashed down safely in the Gulf of Mexico Sunday, completing NASA's first commercial crew, long-duration mission aboard the International Space Station. The return comes nearly six months after the crew members arrived at the microgravity laboratory and also marks the longest-duration mission of a crewed American spacecraft to date.

SpaceX's Crew Dragon, carrying NASA astronauts Michael Hopkins, Victor Glover, and Shannon Walker, and Japan Aerospace Exploration Agency astronaut Soichi Noguchi, returned to Earth in a parachute-assisted splashdown at 2:56 a.m. EDT off the coast of Panama City, Florida. Crews aboard SpaceX recovery vessels successfully recovered the spacecraft and astronauts. After returning to shore, the astronauts will fly back to Houston.

"Welcome home Victor, Michael, Shannon, and Soichi, and congratulations to the teams at NASA and SpaceX who worked so hard to ensure their safe and successful splashdown," said Sen. Bill Nelson, who was confirmed by the Senate to serve as NASA Administrator on April 29. "We've accomplished another incredible spaceflight for America and our commercial and international partners. Safe, reliable transportation to the International Space Station is exactly the vision that NASA had when the agency embarked on the commercial crew program."

NASA's SpaceX Crew-1 mission launched Nov. 15, 2020, on a Falcon 9 rocket from the agency's Kennedy Space Center in Florida. The astronauts named the spacecraft Resilience, in honor of their families, colleagues, and fellow citizens and highlighting the dedication displayed by

the teams involved with the mission and demonstrating that there is no limit to what humans can achieve when they work together. Crew Dragon Resilience docked to the Harmony module's forward port of the space station Nov. 16, nearly 27 hours after liftoff.

Overall, Hopkins, Glover, Walker, and Noguchi traveled 71,242,199 statute miles during their 168 days in orbit (with 167 days aboard the space station), completing 2,688 orbits around Earth. With splashdown, the crew also broke the American crewed spacecraft mission duration record of 84 days, 1 hour, 15 minutes, set by the final Skylab crew in February 1974.

Crew-1 also is the first night splashdown of a U.S. crewed spacecraft since Apollo 8's predawn return in the Pacific Ocean on Dec. 27, 1968, with NASA astronauts Frank Borman, Jim Lovell and Bill Anders.

Throughout their mission, the Crew-1 astronauts contributed to scientific investigations and technology demonstrations, in addition to spacewalks and public engagement events, while aboard the orbiting laboratory. From studying protein crystal development to advance new drug discoveries, to demonstrating robotic assistant technologies, their work advances exploration of the universe while bringing benefits back to Earth.

They also grew crops in both the Advanced Plant Habitat and Veggie plant growth facilities, and conducted tests of a new method for producing semiconductor crystals. The astronauts contributed hundreds of pictures of Earth as part of the Crew Earth Observation investigation, one of the longest-running investigations aboard the space station, which contributes to tracking of natural disasters and changes to our home planet. The crew also tested a new tape dispenser, designed and produced by students as part of the High Schools United with NASA to Create Hardware (HUNCH), during the mission.

On April 5, all four Crew-1 astronauts boarded Resilience for a port relocation maneuver, moving their spacecraft from the forward-facing port to the space-facing port on the Harmony module. The move allowed for the forward-facing port to receive four Crew-2 astronauts upon their arrival to the station April 24. Later this year, SpaceX's 22nd Commercial Resupply Services mission is scheduled to dock at the newly vacant zenith port, bringing with it the first pair of new solar arrays

Eclipsed Moon leaving totality

Rashi Girdhar



Rashi G.

Date: 26 May 2021

Location: Sunnyvale, CA

Imaging Telescope: Orion 5" MAK

Imaging Camera: Phone Camera - Pixel 3XL mounted via Orion Steady Pix Pro cell phone adapter on eyepiece

Mount: Manual Alt-Az

Software: Stock Pixel Camera App



Dates: April 16, 2021, April 17, 2021, April 19, 2021, April 20, 2021, May 1, 2021, May 3, 2021, May 10, 2021, May 12, 2021

Location: Courtyard, Palo Alto, CA

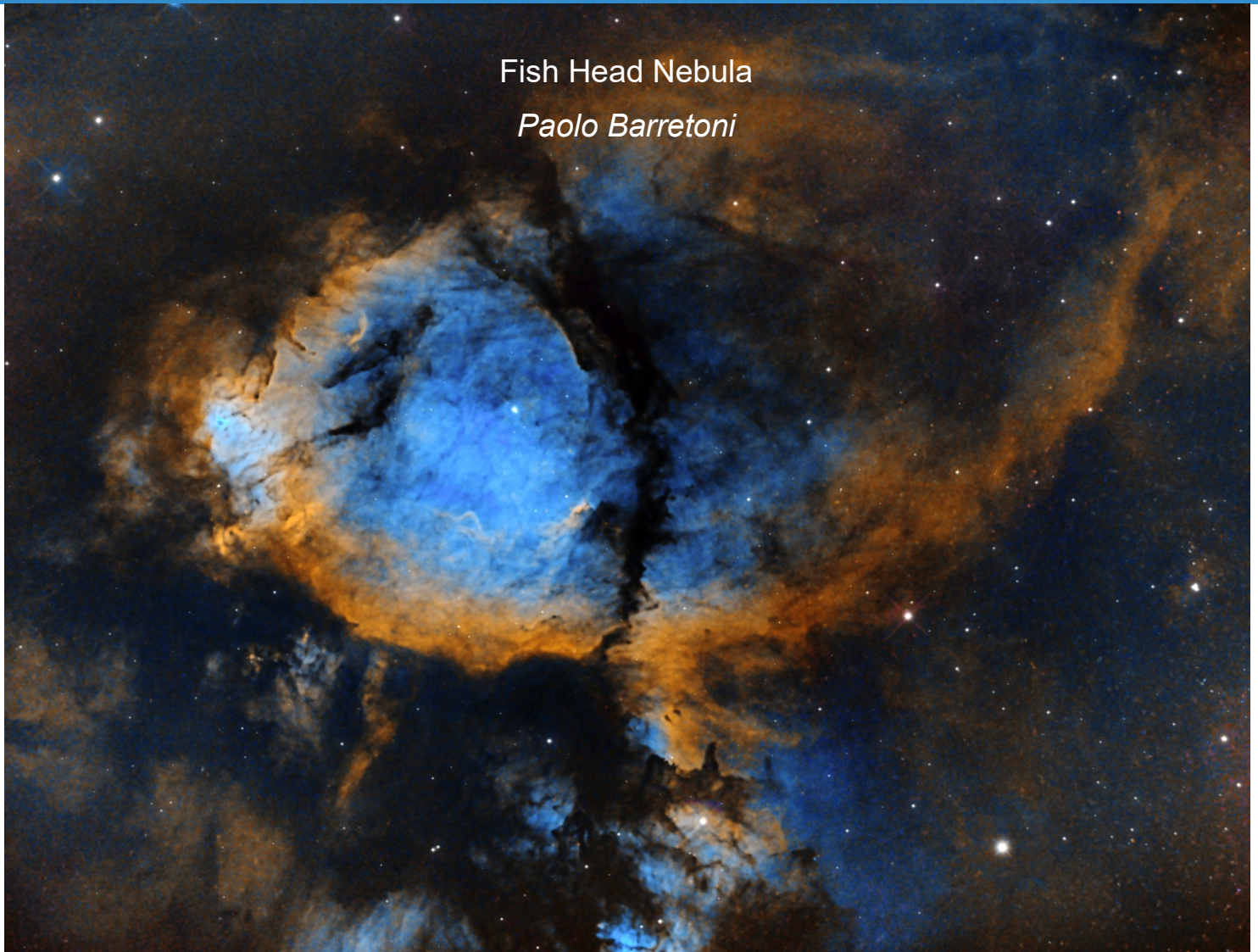
Telescope: TEC 160 FL

Camera: ZWO ASI6200MM Pro

Mount: AP1100GTO

Software: PHD2, PixInsight, Lightroom 6.4, Sequence Generator Pro, Photoshop CS6

SJAA Member Astrophoto Gallery



Fish Head Nebula

Paolo Barretoni

Dates: Sept. 3, 2019, Sept. 15, 2019

Location: Backyard, San Jose, CA

Telescope: Orion 8" Ritchey-Chretien

Camera: ZWO ASI 1600MM Pro

Mount: Orion Atlas AZ/EQ Pro.

Software: Main Sequence Software Sequence Generator Pro 3 · PixInsight

SJAA Member Astrophoto Gallery

Messier 51
PJ Mahany



Dates: April 5 2021

Location: San Jose, CA

Telescope: Stellarvue SV4

Camera: ZWO ASI1600MM-Cool

Mount: Astro-Physics Mach1 GTO

Software: Photoshop, PHD2, Sequence Generator Pro, PixInsight



Kid Spot Jokes:

- ♦ **Why did Mickey Mouse go to outer space?**
To find Pluto.
- ♦ **Why didn't the sun go to college?**
Because it already had a million degrees!
- ♦ **What did the doctor say to the rocketship?**
Time to get your booster shot!

Fun Facts Credit: NASA

Blood moons, Super moons, Blue moons and Harvest moons! Do you know what they are?

You may have heard the terminologies for the different types of full moons, but do you know the actual difference?

Blood Moon:

During a total lunar eclipse, the Earth lines up between the Moon and the Sun. This hides the Moon from the sunlight, and the only light that reaches the Moon's surface is from the edges of the Earth's atmosphere. The air molecules from Earth's atmosphere scatter out most of the blue light. The remaining light reflects onto the Moon's surface with a red glow, making the Moon appear red in the night sky, justifying the appropriate name - Blood Moon!

Super Moon:

A "supermoon" appears to us as a larger-than-usual Moon in our night sky. A supermoon looks larger just because it's a bit closer to Earth. "Supermoon" is actually just a nickname for what astronomers call a perigean full moon – a moon that is full and at its closest point in its orbit around Earth.

Blue Moon:

A Blue Moon is special because it is the "extra" Moon in a season with four full moons. This usually only happens every two-and-a-half years. Since the 1940s, the term "blue moon" has also been used for the second full moon in a calendar month. This usually happens only every two-and-a-half years.

Harvest Moon:

The term "harvest moon" refers to the full, bright Moon that occurs closest to the start of autumn. The name dates from the time before electricity, when farmers depended on the Moon's light to harvest their crops late into the night. The Moon's light was particularly important during fall, when harvests are the largest!



Image of partially eclipsed moon setting behind South Western hills in the early morning (as seen from Sunnyvale, CA)

Credit: Rashi Girdhar

From the Board of Directors

Announcements

- ◇ San Jose Astronomical Association is closely monitoring the coronavirus/COVID-19 situation in Santa Clara County. In accordance with guidance from Santa Clara County Health Department, SJAA leaders have been canceling many events, especially those that involve close contact and shared surfaces on telescopes. Some events, however, have successfully been moved to an online format. Please check our Meetup to determine if an event has been canceled or moved online. The board of directors will periodically review the status of the outbreak and the guidance from the county to determine the impact upon upcoming events.
- ◇ Officer elections for President, Vice President, Treasurer, and Secretary were held during the March board meeting. Rob Jaworski was reelected as Treasurer. Ken Miura was reelected as President. Emi Nikolov was reelected as Secretary. The Vice President position is currently open. If you are interested in taking on this role, please contact secretary@sjaa.net

Board Meeting Excerpts

March 27, 2021

In attendance

Emi, Swami, Sandy, Glenn, Ken, Peter Wolf, Sukhada, Rob J, Vini
Absences: none
Guests: Marianne, Kanch

Officer Elections

Officer elections for President, Vice-President, Treasurer, and Secretary were held during the March board meeting. Rob Jaworski was reelected as Treasurer. Ken Miura was reelected as President. Emi Nikolov was reelected as Secretary. The vice-president position is currently open. If you are interested in taking on this role please contact secretary@sjaa.net. Their terms will be in effect from now until the March board meeting of 2022.

Access to Houge Park

SJAA is working with the city to finalize the terms of the contract, including insurance requirements from the city.

eVscope for SSP Proposal

Proposal to purchase eVscope for the School Star Party program has been approved by the board. The eVscope will be primarily used for the School Star Party program, but might find use during other events too. The board would also like to acknowledge a significant targeted donation to the SJAA to help offset the cost of acquiring the scope. Watch this space for upcoming information!

April 24, 2021

In attendance

Swami, Sandy, Vini, Glenn, Sukhada, Rob J, Vini, Rob J, Ken, Gerry, Peter, Emi
Absences: Wolf
Guests: Marianne, Kanch

Concerns regarding reopening in person events

Re-opening considerations for star parties such as Starry Nights and ITSP were discussed in the meeting. Concerns involving close contacts of the general public with telescopes (e.g. eyepieces) that belong to club members were discussed.

Loaner Program Resurrection

A request letter has been sent to the City of San Jose, but SJAA hasn't received any updates from the city, and still can't access building 1.

Outreach request from Almaden Golf & Country Club

SJAA has recently received an outreach request from Almaden Golf & Country Club asking for a star party and a presentation. A response will be sent, but a date hasn't been set yet.

May 22, 2021

In attendance

Swami, Wolf, Vini, Glenn, Emi, Ken, Sukhada Peter
Absences: Rob J, Sandy
Guests: Manish, Sheena, Karen, Kanch, Joseph, Risham

Volunteer Coordinator

Sheena from Seattle has expressed interest in being a volunteer coordinator. She heard about SJAA via the armchair star party.

Member Communication

Multiple suggestions to try and improve member communication were discussed.

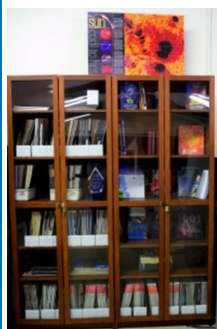
Restarting Events

SJAA had discussions with the Open Space Authority (OSA) to restart some public events in person.

Electronic Assisted Astronomy

Glenn presented a few slides regarding using Electronically Assisted Astronomy (EAA). A proposal and more details to come later.

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick START program is undergoing a redesign to make it more accessible to SJAA members, so signups are currently on hold.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night

sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for

additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Director:	Ken Miura
Vice-President:	Vacant
Treasurer/Dir:	Rob Jaworski
Secretary:	Emi Nikolov
Director:	Vini Carter
Director:	Sandy Mohan
Director:	Wolf Witt
Director:	Swami Nigam
Director:	Peter Melhus
Director:	Glenn Newell
Director:	Sukhada Palav
Ephemeris Newsletter -	
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Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
In-Town Star Party:	Muditha Kanchana
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Beth Johnson
Quick START	Dave Ittner
Donations:	Dave Ittner*
Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Cat Cvengros
Refreshments:	Marianne Damon
Social Media Manager:	Beth Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

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San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

☐ **New** ☐ **Renewal**

☐ **\$20** Regular Membership with online Ephemeris

☐ **\$30** Regular Membership with hardcopy Ephemeris mailed to below address

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

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